Fiscal Year:	FY 2019	**Task Last Updated:**	FY 07/30/2019
PI Name:	Hada, Megumi Ph.D.		
Project Title:	Computational Model Prediction and Biological Validation using Simplified Mixed Field Exposures for the Development of a GCR Reference Field		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	mehada@pvamu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	936-261-3155
Organization Name:	Prairie View A&M University		
PI Address 1:	College of Arts and Sciences, PO BOX 519, MS-2230, New Science Bldg		
PI Address 2:			
PI Web Page:			
City:	Prairie View	**State:**	TX
Zip Code:	77446	**Congressional District:**	10
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-RADIATION. Appendix D: Ground-Based Studies in Space Radiobiology
Start Date:	08/26/2016	**End Date:**	03/31/2021
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 3/31/2021 per NSSC information (Ed., 5/28/20) NOTE: End date changed to 3/31/2020 per NSSC information (Ed., 8/8/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Patel, Zarana Ph.D. (Wyle Laboratories, Inc.) Plante, Ianik Ph.D. (Wyle Laboratories, Inc.) Ponomarev, Artem Ph.D. (Wyle Laboratories, Inc.) Shavers, Mark Ph.D. (Wyle Laboratories, Inc.) Slaba, Tony Ph.D. (NASA Langley Research Center)		
Grant/Contract No.:	NNX16AR97G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This work addresses the need to develop and utilize mixed field irradiation protocols that approximately represent the shielded tissue environment in space and that can be combined with single-beam studies to validate and further improve cancer risk models. Extensive studies have been carried out on the induction of chromosomal aberrations by low- and high-LET (linear energy transfer) radiation in human lymphocytes, fibroblasts, and epithelial cells exposed in vitro. The results, which are consistent with computational modeling predictions, establish chromosome aberration models as an LET-sensitive tool for predicting damage from single ion exposures. However, there is a lack of data on chromosome aberrations induced by low dose rate chronic exposure and mixed field beams such as those expected in space. These proposed studies will define a simplified mixed field and make model predictions of the effects of dose rate, mixed fields, and shielding on expected biological damage. Chromosome aberration studies at NASA Space Radiation Laboratory (NSRL) will provide the biological validation needed to extend the computational models over a broader range of experimental conditions ((more complicated mixed fields leading up to the galactic cosmic radiation (GCR) simulator)), helping to reduce uncertainties in radiation quality effects and dose-rate dependence in cancer risk models (Cancer-3 and -4). These models can then be used to answer some of the open questions regarding requirements for a full GCR reference field, including particle type and number, energy, dose rate, and delivery order.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This strategy of single and mixed field exposures and varying dose rates along with a well-defined biological assay and established computational models will provide a unique opportunity to develop an intermediate mixed field that can closely approximate the GCR reference field. This represents a critical step towards using the GCR simulator at NSRL to reduce uncertainties in radiation quality effects and dose-rate dependence in cancer risk models.
Task Progress:	Experimental Analysis of chromosome aberrations: Human normal fibroblast cells and blood were irradiated with single or mixed beam of silicon, titanium, and iron-ions without shielding in NSRL 19A run at Brookhaven National Laboratory. Beams, energy, and dose used are as follows: Single beam: Si (260 MeV/n) 10 cGy, 20 cGy ; Ti (1000 MeV/n) 10 cGy, 20 cGy ; Fe (600 MeV/n) 10 cGy, 20 cGy Mixed beam: SI (260 MeV/n) + Ti (1000 MeV/n) + Fe (600 MeV/n); 20 cGy + 20 cGy + 20 cGy, 10 cGy + 10 cGy + 10 cGy Chromosomes were collected in the first mitosis after exposure (total 18 samples). Analysis of chromosome aberrations (CA) with 3-color fluorescent in situ hybridization (FISH) chromosome painting methods has been progressing. Analysis of chromosome samples collected year 1 and year 2 were continued and total 150 samples analysis were completed. 4 beam mix and 6 beam mix exposure results showed that 1) Both 4 beam mix and 6 beam mixed exposure shows less than additive effects in chromosome aberration frequency both in human fibroblasts and blood lymphocytes. 2) 4 beam mix and 6 beam mix showed similar frequency of chromosome aberrations. Modeling We have created new simulation program RITCARD (Radiation induced tracks, chromosome aberrations, repair, and damage) in year 2. The code is largely using C++ objects named classes, and mostly follows coding standards. In year 3, we have implemented the bi-exponential repair kinetics model into the RITCARD, thereby replacing the previous simplified repair kinetics approach carried over from the NASA Radiation Track Image (NASARTI) program. RITCARD simulates the geometric configuration of chromosomes by random walk, the radiation-induced breaks and their repair, and the assessment of various types of CAs. The implementation of the bi-exponential repair relies on a mathematically exact algorithm. In the model, the complexity of the breaks or free end pairs determines whether the breaks are repaired via the fast or slow repair pathway. As we assume that higher local energy deposition is necessary to create a complex break, a dose threshold to determine whether a break is complex or simple was used. Improper rejoinings (those not originating from the same break) are considered complex repairs. The probabilities of the free ends pairs to repair are weighted by the distance of the free ends as exp(-distance2/s2). By using the algorithm, we have found that the fraction of unrepaired breaks increases with linear energy transfer (LET), that the repair kinetics curves are independent on the dose, and that the dose-response curves for simple and complex exchanges are well approximated by a linear-quadratic curve. A peak in the alpha coefficient of the dose-response curve at about 100 keV/µm is also found, evocating a relative biological effectiveness (RBE) peak.
Bibliography Type:	Description: (Last Updated: 06/26/2025)
Abstracts for Journals and Proceedings	Plante I, Ponomarev A, Patel Z, Slaba T, Hada M. "Simulation of radiation-induced chromosome aberrations." Presented at 64th Annual Meeting of the Radiation Research Society, Chicago, IL, September 23-26, 2018. RRS2018 Abstract Book. 64th Annual Meeting of the Radiation Research Society, Chicago, IL, September 23-26, 2018. , Sep-2018
Abstracts for Journals and Proceedings	Hada M, Rhone J, Beitman A, Plante I, Ponomarev A, Slaba T, Patel Z, Saganti P. "Computational model prediction and biological validation using simplified mixed field exposures for the development of GCR reference field." Presented at 64th Annual Meeting of the Radiation Research Society, Chicago, IL, September 23-26, 2018. RRS2018 Abstract Book. 64th Annual Meeting of the Radiation Research Society, Chicago, IL, September 23-26, 2018. , Sep-2018

Abstracts for Journals and Proceedings	Ponomarev AL, Plante I, Slaba T, Patel Z, Hada M. "Application of multi-scale methodology to modeling the effects of space radiation: heavy ion tracks, random walk nuclei models, repair of chromatin breaks and chromosomal aberrations." Presented at 64th Annual Meeting of the Radiation Research Society, Chicago, IL, September 23-26, 2018. RRS2018 Abstract Book. 64th Annual Meeting of the Radiation Research Society, Chicago, IL, September 23-26, 2018. , Sep-2018
Abstracts for Journals and Proceedings	Hada M, Rhone J, Beitman A, Plante I, Ponomarev A, Slaba T, Patel Z, Saganti P. "Computational model prediction and biological validation using simplified mixed field exposures for the development of a GCR reference field." Presented at 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. NASA 2019 HRP-IWS Abstract Book. 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. , Jan-2019
Abstracts for Journals and Proceedings	Plante I, Ponomarev A, Patel ZS, Slaba T, Hada M. "RITCARD: Radiation induced tracks, chromosome aberrations, repair, and damage." Presented at 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. NASA 2019 HRP-IWS Abstract Book. 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. , Jan-2019
Abstracts for Journals and Proceedings	Ponomarev AL, Plante I, Patel ZS, Slaba T, Hada M. "Computer visualization of the experimental count of chromosomal aberrations." Presented at 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. NASA 2019 HRP-IWS Abstract Book. 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. , Jan-2019
Abstracts for Journals and Proceedings	Beitman AJ, Rhone JR, Chang PY, Blakely EA, Sachs RK, Saganti PB, Hada M. "Chromosome aberrations induced by GCR simulated mixed beam exposure in human lymphocytes, fibroblasts and epithelial cells." Presented at 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. NASA 2019 HRP-IWS Abstract Book. 2019 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2019. , Jan-2019
Abstracts for Journals and Proceedings	Hada M. "Chromosome aberrations induced by GCR simulated mixed beam exposure in human lymphocytes, fibroblasts and epithelial cells." Presented at 8th Heavy Ion Beam Medical Engineering Collaboration Seminar, Maebashi, Japan, March 13, 2019. Abstract Book. 8th Heavy Ion Beam Medical Engineering Collaboration Seminar, Maebashi, Japan, March 13, 2019. , Mar-2019
Articles in Peer-reviewed Journals	Plante I, Ponomarev A, Patel Z, Slaba T, Hada M. "RITCARD: Radiation induced tracks, chromosome aberrations, repair, and damage." Radiat Res. 2019 Sep;192(3):282-98. Epub 2019 Jul 11. https://doi.org/10.1667/RR15250.1 ; PubMed PMID: 31295089 , Sep-2019
NASA Technical Documents	Plante I, Ponomarev A, Patel Z, Slaba T, Hada M. "RITCARD: Radiation induced tracks, chromosome aberrations, repair, and damage." NASA Disclosure of Invention and New Technology (Software)(MSC-26560-1), October 2018. , Oct-2018